SCH3U Final Exam Chemistry Test Guide

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How To Use This Guide

Use this as a print-ready final exam review package. Complete the topic checklist first, then try the practice test without notes. Mark your work using the answer key and redo any question where your reasoning was incomplete, even if the final answer was correct.

High-Yield Topic Checklist

- Chemical naming: acids, ionic compounds, polyatomic ions, and hydrates.
- Mole calculations: molar mass, mass-to-moles, particles-to-moles, and limiting reagents.
- Stoichiometry: balanced equations, theoretical yield, actual yield, and percent yield.
- Solutions: dissociation, concentration, dilution, saturation, solubility, and precipitation.
- Net ionic equations: spectator ions, precipitates, and single displacement reactions.
- Acids and bases: pH, strong versus weak acids, neutralization, titration vocabulary, and indicators.
- Gases: pressure, volume, temperature, Boyle's law, Charles's law, and kinetic molecular theory.
- Water treatment: primary treatment, secondary treatment, filtration, disinfection, and common pollutants.

Formula And Reference Sheet

- n = m / M
- m = nM
- C = n / V
- n = CV
- C1V1 = C2V2
- percent yield = actual yield / theoretical yield x 100%
- Boyle's law: P1V1 = P2V2 at constant temperature
- Charles's law: V1 / T1 = V2 / T2, where temperature is in kelvin
- 0 degrees C = 273 K
- A saturated solution contains the maximum amount of dissolved solute at a stated temperature.
- A precipitate forms when ions combine to produce an insoluble compound.

Section A: Multiple Choice

Choose the best answer.

1. What is the correct name for HCl(aq)?
    A. hydrogen chloride
    B. hydrochloric acid
    C. chloric acid
    D. hypochlorous acid
    E. aqueous chlorine

2. What mass is present in 0.250 mol of CaCO3?
    A. 10.0 g
    B. 25.0 g
    C. 40.1 g
    D. 50.0 g
    E. 100.1 g

3. In a chemical reaction, the limiting reagent is the reactant that
    A. has the smallest mass.
    B. is present as a liquid.
    C. is left over after the reaction.
    D. is completely consumed first.
    E. is produced in the smallest amount.

4. Which balanced equation represents the formation of magnesium nitride?
    A. Mg + N2 -> MgN2
    B. 3Mg + N2 -> Mg3N2
    C. Mg3 + N2 -> Mg3N2
    D. 2Mg + N2 -> 2MgN
    E. Mg + 3N -> Mg3N2

5. If 6.00 g of Mg reacts with 6.00 g of O2 according to 2Mg + O2 -> 2MgO, which reactant is limiting?
    A. Mg
    B. O2
    C. MgO
    D. both reactants equally
    E. not enough information

6. A student obtains 12.0 g of product when the theoretical yield is 15.0 g. What is the percent yield?
    A. 60.0%
    B. 75.0%
    C. 80.0%
    D. 88.0%
    E. 125%

7. Which mixture is not a solution?

A. salt water
B. oxygen dissolved in water
C. brass
D. sand and water
E. vinegar

8. The process in which an ionic compound separates into ions in water is called

A. combustion.
B. neutralization.
C. dissociation.
D. condensation.
E. filtration.

9. Which dissociation equation is correct for aluminum nitrate?

A. Al(NO3)3(s) -> Al3+(aq) + 3NO3-(aq)
B. Al(NO3)3(s) -> Al+(aq) + NO3-(aq)
C. AlNO3(s) -> Al3+(aq) + NO3-(aq)
D. Al(NO3)3(s) -> 3Al+(aq) + NO3-3(aq)
E. Al(NO3)3(s) -> Al3+(aq) + NO3-3(aq)

10. What final volume is needed to dilute 250 mL of 2.0 mol/L NaCl to 0.50 mol/L?

A. 125 mL
B. 250 mL
C. 500 mL
D. 750 mL
E. 1000 mL

11. A solution that can dissolve more solute at the same temperature is

A. saturated.
B. unsaturated.
C. supersaturated.
D. concentrated.
E. neutralized.

12. Which ion would form a precipitate with Pb2+(aq) according to common solubility rules?

A. NO3-
B. Na+
C. Cl-
D. K+
E. NH4+

13. What is the correct net ionic equation when zinc metal is placed into copper(II) sulfate solution?
    A. Zn(s) + Cu2+(aq) -> Zn2+(aq) + Cu(s)
    B. Zn2+(aq) + Cu(s) -> Zn(s) + Cu2+(aq)
    C. Zn(s) + SO4 2-(aq) -> ZnSO4(s)
    D. Cu2+(aq) + SO4 2-(aq) -> CuSO4(s)
    E. Zn(s) + CuSO4(aq) -> ZnSO4(aq) + Cu(s)

14. Which compound is most likely insoluble in water?
    A. NaNO3
    B. KCl
    C. NH4NO3
    D. AgCl
    E. Li2SO4

15. Which dissociation equation is correct for barium hydroxide?
    A. BaOH(s) -> Ba+(aq) + OH-(aq)
    B. Ba(OH)2(s) -> Ba2+(aq) + 2OH-(aq)
    C. Ba(OH)2(s) -> 2Ba+(aq) + 2OH-(aq)
    D. Ba(OH)2(s) -> Ba2+(aq) + OH2-(aq)
    E. Ba(OH)3(s) -> Ba3+(aq) + 3OH-(aq)

16. Which statement is not a typical property of bases?
    A. They feel slippery.
    B. They turn red litmus blue.
    C. They have pH values greater than 7.
    D. They neutralize acids.
    E. They produce H+(aq) as their main ion in water.

17. Which statement best compares 0.10 mol/L HCl and 0.10 mol/L CH3COOH?
    A. HCl is a strong acid and ionizes almost completely.
    B. CH3COOH is a strong acid and ionizes almost completely.
    C. Both acids produce the same hydronium concentration.
    D. Neither acid conducts electricity.
    E. Both solutions are basic.

18. The careful addition of a solution from a buret to determine concentration is called
    A. titration.
    B. filtration.
    C. distillation.
    D. chromatography.
    E. evaporation.

19. A liquid has
    A. definite shape and definite volume.
    B. no definite shape and definite volume.
    C. definite shape and no definite volume.
    D. no definite shape and no definite volume.
    E. no particles.

20. Gas pressure results from
    A. gas molecules attracting each other strongly.
    B. gas molecules colliding with container walls.
    C. liquid particles evaporating.
    D. the mass of the container.
    E. dissolved ions separating.

21. If the pressure on a gas doubles at constant temperature, its volume
    A. doubles.
    B. triples.
    C. halves.
    D. remains constant.
    E. becomes zero.

22. Which stage of wastewater treatment removes large suspended solids by screening and settling?
    A. primary treatment
    B. secondary treatment
    C. tertiary treatment
    D. disinfection
    E. softening

23. Which acid name matches $HNO_3(aq)$?
    A. nitrous acid
    B. nitric acid
    C. hydronitric acid
    D. nitrate acid
    E. nitrogenous acid

24. What volume of 0.200 mol/L HCl contains 0.0500 mol of HCl?
    A. 0.0100 L
    B. 0.0250 L
    C. 0.100 L
    D. 0.250 L
    E. 4.00 L

25. In a precipitation reaction, spectator ions are ions that

A. form the solid product.

B. change oxidation state.

C. appear unchanged before and after the reaction.

D. are always hydrogen and hydroxide ions.

E. evaporate from the solution.

Section B: Short Answer And Calculations

Show your work for calculation questions.

26. Calculate the molar mass of K_2SO_4.

27. How many moles are in 18.0 g of H_2O?

28. Balance the equation: $C_3H_8 + O_2 \rightarrow CO_2 + H_2O$.

29. A 0.750 L solution contains 0.300 mol of NaOH. Calculate its concentration.

30. How much water must be added to 200 mL of 1.50 mol/L KBr to make a 0.500 mol/L solution?

31. Write the total ionic equation and net ionic equation for $AgNO_3(aq) + NaCl(aq) \rightarrow AgCl(s) + NaNO_3(aq)$.

32. Explain why a saturated solution can still contain undissolved solid at the bottom of the container.

33. A gas has a volume of 3.00 L at 100 kPa. What is its volume at 150 kPa if temperature is constant?

34. A student titrates 25.00 mL of HCl with 0.100 mol/L NaOH. The endpoint occurs after 18.60 mL of NaOH is added. Calculate the concentration of HCl.

35. Name two differences between strong acids and weak acids.

Section C: Extended Response

36. Limiting reagent and percent yield: Aluminum reacts with chlorine gas according to $2Al + 3Cl_2 \rightarrow 2AlCl_3$. If 5.40 g of Al reacts with 20.0 g of Cl_2, determine the limiting reagent, theoretical yield of $AlCl_3$, and percent yield if 22.5 g of $AlCl_3$ is collected.

37. Solutions and precipitation: A student mixes 50.0 mL of 0.200 mol/L $BaCl_2$ with 50.0 mL of 0.200 mol/L Na_2SO_4. Predict whether a precipitate forms, identify it, write the net ionic equation, and calculate the moles of precipitate expected.

38. Gas laws: A sealed syringe contains 45.0 mL of gas at 22 degrees C. The syringe is warmed to 60 degrees C at constant pressure. Calculate the new volume and explain why the volume changes.

39. Water treatment: Describe the purpose of primary treatment, secondary treatment, filtration, and disinfection. For each stage, identify one type of contaminant or risk it helps reduce.

40. Acid-base thinking: A student claims that any acid with the same concentration must have the same pH. Use HCl and acetic acid as examples to explain why this claim is not correct.

Answer Key

1. B
2. B
3. D
4. B
5. A
6. C
7. D
8. C
9. A
10. E
11. B
12. C
13. A
14. D
15. B
16. E
17. A
18. A
19. B
20. B
21. C
22. A
23. B
24. D
25. C

26. K_2SO_4 = 2(39.10) + 32.07 + 4(16.00) = 174.27 g/mol.

27. n = 18.0 g / 18.02 g/mol = 0.999 mol, which is about 1.00 mol.

28. $C_3H_8 + 5O_2 \rightarrow 3CO_2 + 4H_2O$.

29. C = n / V = 0.300 mol / 0.750 L = 0.400 mol/L.

30. $C_1V_1 = C_2V_2$. (1.50)(200 mL) = (0.500)(V_2), so V_2 = 600 mL. Water added = 600 mL - 200 mL =

400 mL.

31. Total ionic: Ag+(aq) + NO3-(aq) + Na+(aq) + Cl-(aq) -> AgCl(s) + Na+(aq) + NO3-(aq). Net ionic: Ag+(aq) + Cl-(aq) -> AgCl(s).

32. At saturation, the solution holds the maximum dissolved amount at that temperature. Extra solute remains undissolved because the dissolving and crystallizing processes are balanced.

33. P1V1 = P2V2. (100 kPa)(3.00 L) = (150 kPa)(V2), so V2 = 2.00 L.

34. HCl + NaOH -> NaCl + H2O. n NaOH = (0.100 mol/L)(0.01860 L) = 0.001860 mol. The mole ratio is 1:1, so n HCl = 0.001860 mol. C HCl = 0.001860 mol / 0.02500 L = 0.0744 mol/L.

35. Strong acids ionize almost completely in water; weak acids ionize only partially. At the same concentration, strong acids usually produce higher hydronium concentration, lower pH, and greater conductivity.

36. Moles Al = 5.40 g / 26.98 g/mol = 0.200 mol. Moles Cl2 = 20.0 g / 70.90 g/mol = 0.282 mol. For 0.200 mol Al, required Cl2 = 0.300 mol, so Cl2 is limiting. Moles AlCl3 = 0.282 mol Cl2 x 2 mol AlCl3 / 3 mol Cl2 = 0.188 mol. Mass AlCl3 = 0.188 mol x 133.33 g/mol = 25.1 g. Percent yield = 22.5 g / 25.1 g x 100% = 89.6%.

37. BaSO4 precipitates because sulfates are generally soluble except with Ba2+, Pb2+, Sr2+, and Ca2+ to varying degrees. Net ionic: Ba2+(aq) + SO4 2-(aq) -> BaSO4(s). Moles Ba2+ = 0.0500 L x 0.200 mol/L = 0.0100 mol. Moles SO4 2- = 0.0500 L x 0.200 mol/L = 0.0100 mol. Expected precipitate = 0.0100 mol BaSO4.

38. Convert temperatures to kelvin: T1 = 295 K and T2 = 333 K. V2 = V1T2 / T1 = 45.0 mL x 333 K / 295 K = 50.8 mL. Volume increases because gas particles have greater kinetic energy at higher temperature and occupy more volume at constant pressure.

39. Primary treatment removes large solids and settleable particles through screening and settling. Secondary treatment uses biological processes to reduce dissolved and suspended organic matter. Filtration removes fine particles that remain after earlier steps. Disinfection reduces disease-causing microorganisms before water is released or reused.

40. The claim is incorrect because acid concentration is not the same as degree of ionization. HCl is a strong acid and ionizes almost completely, producing a high H3O+ concentration. Acetic acid is weak and ionizes only partially, so at the same formal concentration it produces less H3O+ and has a higher pH than HCl.

Final Study Advice

- Redo all mole and solution calculations with units written at every step.
- Memorize common solubility exceptions, especially compounds of Ag+, Pb2+, Ba2+, carbonate,

phosphate, and hydroxide.
- For net ionic equations, always write states first, split strong aqueous electrolytes into ions, then cancel spectators.
- For acid-base questions, separate concentration from strength.
- For gas laws, convert Celsius to kelvin before using temperature in a formula.